

BIRCH (Balanced Iterative Reducing and Clustering Using Hierarchies)

- BIRCH is designed for clustering a large amount of numeric data by integrating hierarchical clustering and other clustering methods such as iterative partitioning.
- It overcomes the two difficulties in agglomerative clustering methods:
 - Scalability
 - The inability to undo what was done in the previous step.
- BIRCH uses the notions of *clustering feature* to summarize a cluster, and *clustering feature tree (CF-tree)* to represent a cluster hierarchy.

Clustering Feature Vector in BIRCH

- Consider a cluster of n d -dimensional data objects or points. The **clustering feature (CF)** of the cluster is a 3-D vector summarizing information about clusters of objects.
- It is defined as **Clustering Feature (CF)**: $CF = (N, LS, SS)$

N : Number of data points

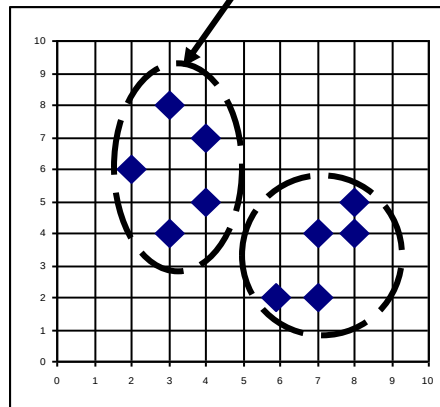
LS : linear sum of N points:

$$\sum_{i=1}^N X_i$$

SS : square sum of N points

$$\sum_{i=1}^N X_i^2$$

$CF = (5, (16,30),(54,190))$



(3,4)

(2,6)

(4,5)

(4,7)

(3,8)

Clustering Feature Vector in BIRCH

- Using a clustering feature, we can easily derive many useful statistics of a cluster.
- For example, the cluster's centroid, $\mathbf{x}_0$, radius, R , and diameter, D , are

$$\mathbf{x}_0 = \frac{\sum_{i=1}^n \mathbf{x}_i}{n} = \frac{LS}{n},$$

$$R = \sqrt{\frac{\sum_{i=1}^n (\mathbf{x}_i - \mathbf{x}_0)^2}{n}} = \sqrt{\frac{nSS - 2LS^2 + nLS}{n^2}}$$

$$D = \sqrt{\frac{\sum_{i=1}^n \sum_{j=1}^n (\mathbf{x}_i - \mathbf{x}_j)^2}{n(n-1)}} = \sqrt{\frac{2nSS - 2LS^2}{n(n-1)}}.$$

Clustering Feature Vector in BIRCH

- Summarizing a cluster using the clustering feature can avoid storing the detailed information about individual objects or points.
- We only need a constant size of space to store the clustering feature.
- This is the key to BIRCH efficiency in space.
- Clustering features are *additive*.
 - E.g. for two disjoint clusters, C_1 and C_2 , with the clustering features $CF_1 = (n_1, LS_1, SS_1)$ and $CF_2 = (n_2, LS_2, SS_2)$, respectively, the clustering feature for the cluster that formed by merging C_1 and C_2 is simply

$$CF_1 + CF_2 = \langle n_1 + n_2, LS_1 + LS_2, SS_1 + SS_2 \rangle$$

CF-Tree in BIRCH

- A CF tree is a height-balanced tree that stores the clustering features for a hierarchical clustering
 - A nonleaf node in a tree has descendants or “children”
 - The nonleaf nodes store sums of the CFs of their children
- A CF tree has two parameters
 - Branching factor: max # of children
 - Threshold: max diameter of sub-clusters stored at the leaf nodes

The CF Tree Structure

Root

$B = 7$

$L = 6$

CF ₁	CF ₂	CF ₃		CF ₆
child ₁	child ₂	child ₃		child ₆

Non-leaf node

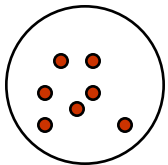
CF ₁	CF ₂	CF ₃		CF ₅
child ₁	child ₂	child ₃		child ₅

Leaf node

Leaf node

prev	CF ₁	CF ₂		CF ₆	next
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prev	CF ₁	CF ₂		CF ₄	next
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BIRCH

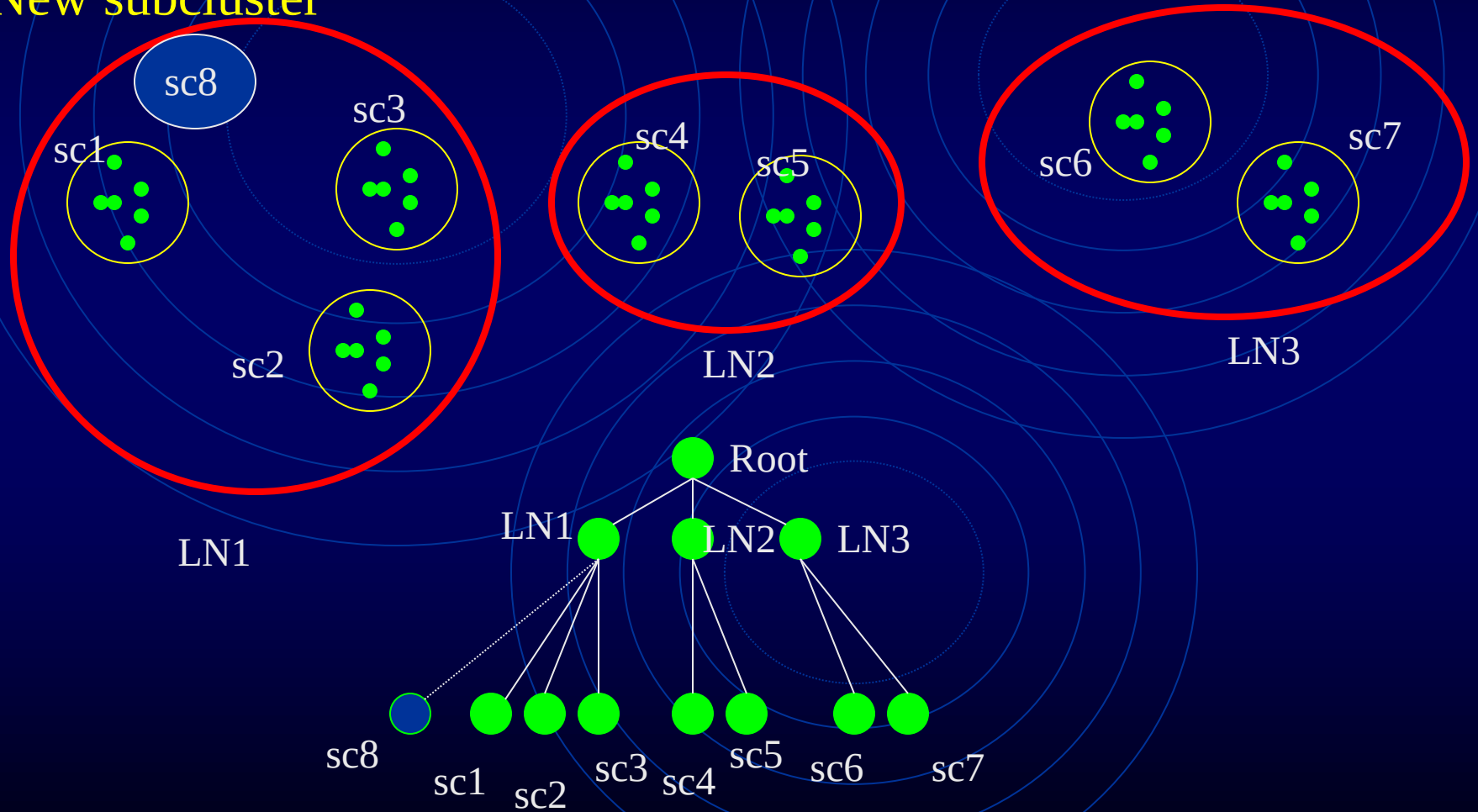
- Incrementally construct a CF (Clustering Feature) tree, a hierarchical data structure for multiphase clustering
- The primary phases are
 - Phase 1: scan DB to build an initial in-memory CF tree (a multi-level compression of the data that tries to preserve the inherent clustering structure of the data)
 - Phase 2: use an arbitrary clustering algorithm to cluster the leaf nodes of the CF-tree
- *Scales linearly*: finds a good clustering with a single scan and improves the quality with a few additional scans
- *Weakness*: handles only numeric data, and sensitive to the order of the data record

The Birch Algorithm

- For each point in the input
 - Find closest leaf entry
 - Add point to leaf entry and update CF
 - If entry diameter $>$ max_diameter, then split leaf, and possibly parents
 - Information about the object is passed toward the root of the tree.
- The size of the CF-tree can be changed by modifying the threshold.
- Algorithm is $O(n)$
- Concerns
 - Sensitive to insertion order of data points
 - Since we fix the size of leaf nodes, so clusters may not be so natural
 - Clusters tend to be spherical given the radius and diameter measures

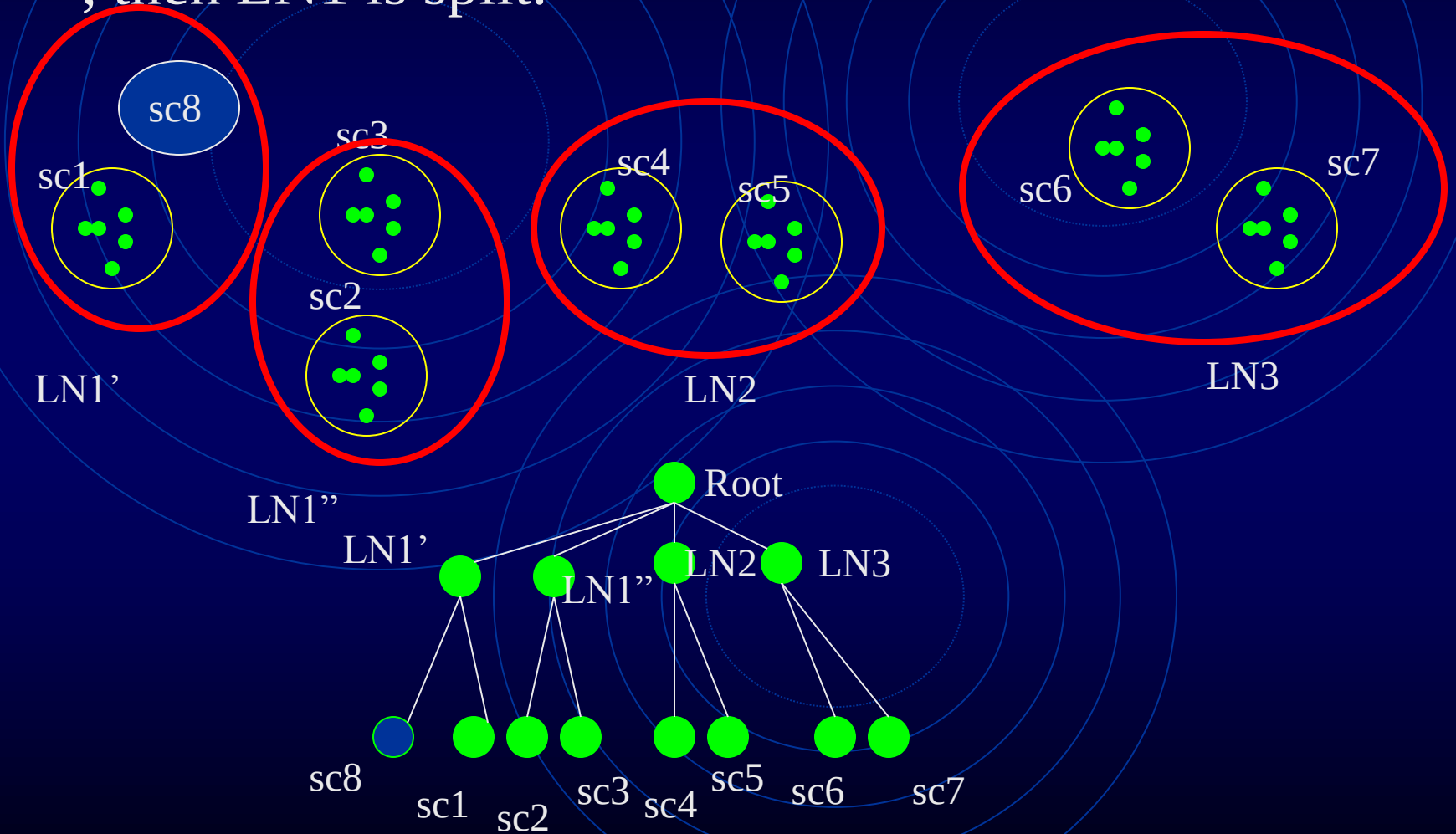
Example of the BIRCH Algorithm

New subcluster



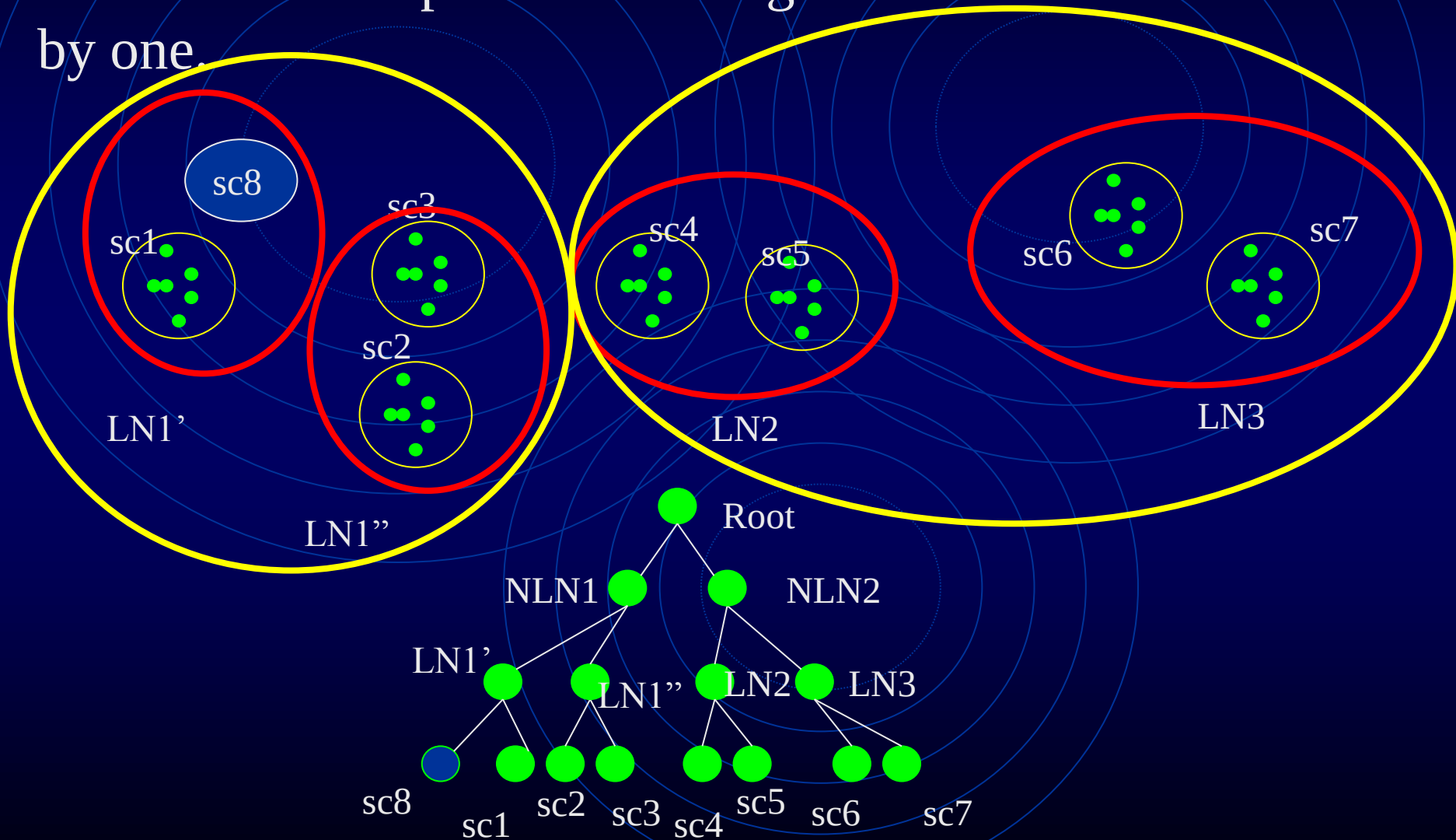
Merge Operation in BIRCH

If the branching factor of a leaf node can not exceed 3, then LN1 is split.



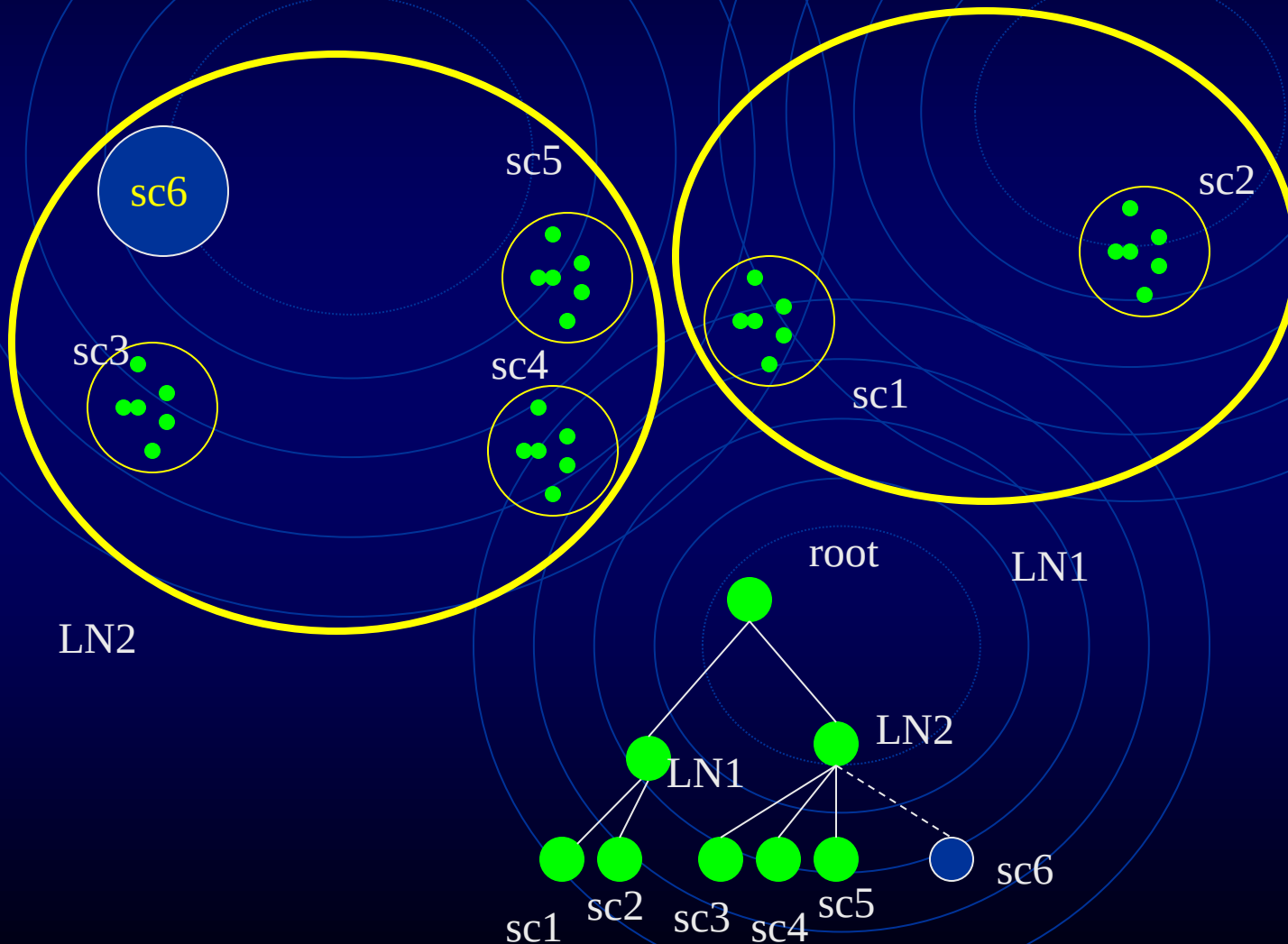
Merge Operation in BIRCH

If the branching factor of a non-leaf node can not exceed 3, then the root is split and the height of the CF Tree increases by one.

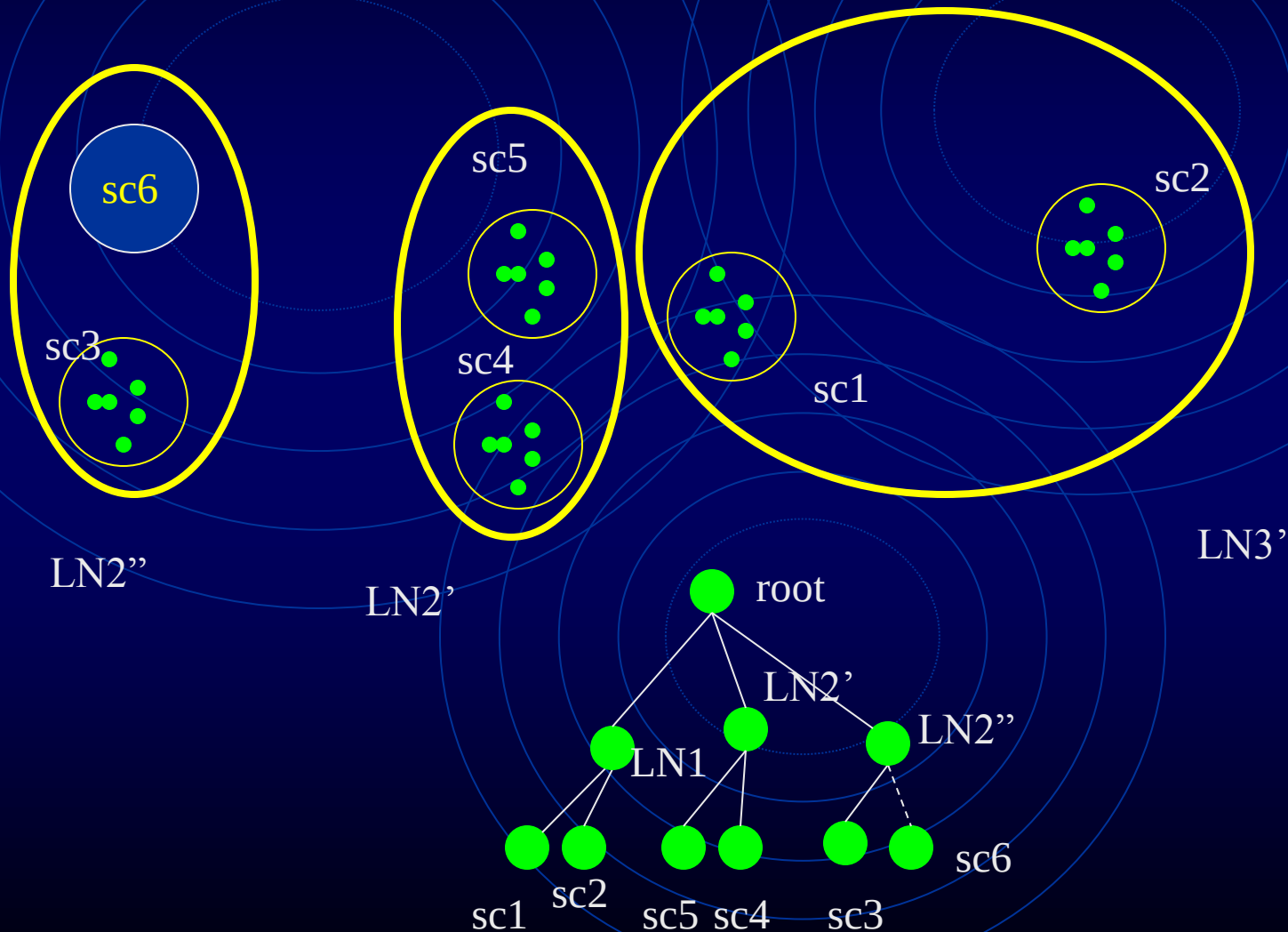


Merge Operation in BIRCH

Assume that the subclusters are numbered according to the order of formation.



If the branching factor of a leaf node can not exceed 3,
Then LN2 is split.



LN2' and LN1 will be merged, and the newly formed Node will be split immediately.

